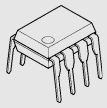
DUAL OPERATIONAL AMPLIFIER

■ GENERAL DESCRIPTION

The NJM4558 is a dual high-gain operational amplifier with internal compensation circuit and constructed on a single silicon chip. It offers excellent characteristics by combining the parameters adjusted for a monolithic chip. The channel separation characteristic is suitable for measuring instruments.

■ PACKAGE OUTLINE

Discontinued



NJM4558D
(DIP8)

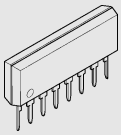


NJM4558M
(DMP8)



NJM4558V
(SSOP8)

Discontinued



NJM4558L
(SIP8)



NJM4558E
(SOP8)

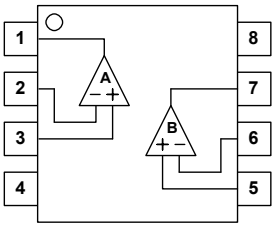
■ FEATURES

- Operating Voltage (±4V~±18V)
- High Voltage Gain (100dB typ.)
- High Input Resistance (5MΩ typ.)
- Bipolar Technology
- Package Outline DIP8 (Discontinued), DMP8, SSOP8, SIP8 (Discontinued) SOP8 JEDEC 150mil

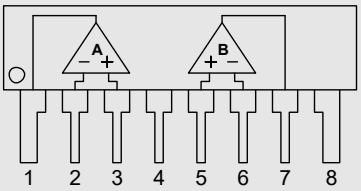
	Equivalent Input Noise Voltage V_{NI}
NJM4558M / NJM4558V / NJM4558E / NJM4558D (Discontinued) / NJM4558L (Discontinued)	Typ. 1.4 μVrms
NJM4558M-D / NJM4558E-D / NJM4558D-D (Discontinued) / NJM4558L-D (Discontinued)	Max. 1.8μVrms

■ PIN CONFIGURATION

(Top View)



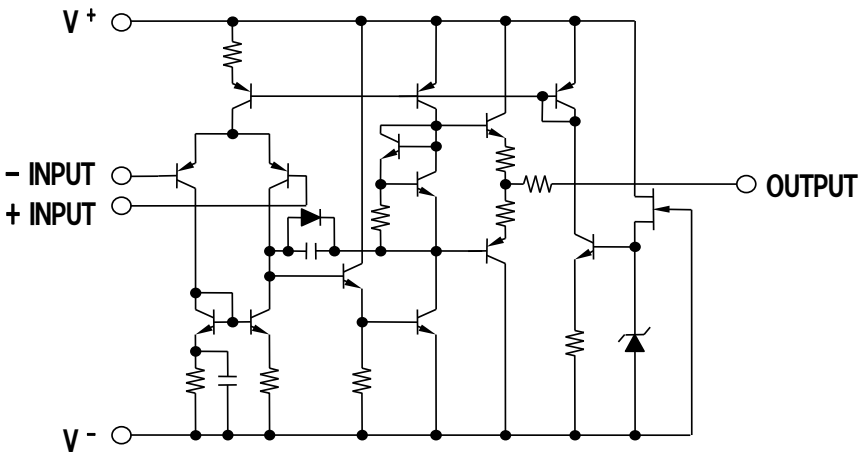
NJM4558M, NJM4558E, NJM4558V
NJM4558D (Discontinued)



NJM4558L (Discontinued)

PIN FUNCTION
1. A OUTPUT
2. A - INPUT
3. A +INPUT
4. V^-
5. B +INPUT
6. B - INPUT
7. B OUTPUT
8. V^+

■ EQUIVALENT CIRCUIT (1/2 Shown)



■ ABSOLUTE MAXIMUM RATINGS

(Ta=25˚C)

PARAMETER	SYMBOL	RATINGS	UNIT
Supply Voltage	V^+/V^-	± 18	V
Differential Input Voltage	V_{ID}	± 30	V
Input Voltage	V_{IC}	± 15 (note1)	V
Power Dissipation *1	P_D	(DIP8) 500 (DMP8) 300 (SOP8) 300 (SSOP8) 250 (SIP8) 800	mW
Operating Temperature Range	T_{opr}	-40~+85	˚C
Storage Temperature Range	T_{stg}	-40~+125	˚C

(note1) For supply voltage less than ±15V,the absolute maximum input voltage is equal to the supply voltage.

*1) IC alone

■ ELECTRICAL CHARACTERISTICS

(V^+/V^-=±15V,Ta=25˚C)

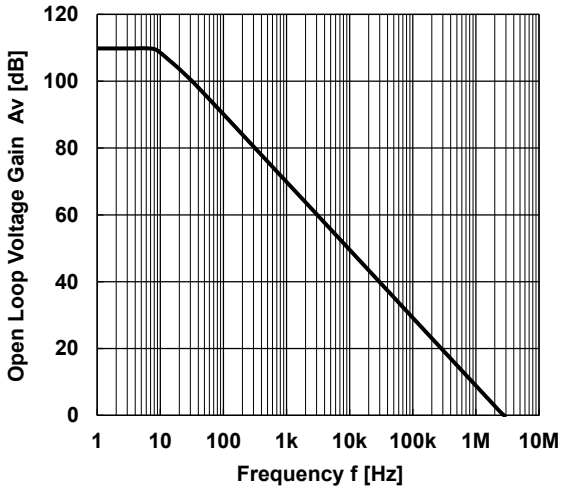
PARAMETER	SYMBOL	TEST CONDITION		MIN.	TYP.	MAX.	UNIT
Input Offset Voltage	V_{IO}	R_S≤10kΩ		-	0.5	6	mV
Input Offset Current	I_{IO}			-	5	200	nA
Input Bias Current	I_B			-	25	500	nA
Input Resistance	R_{IN}			0.3	5	-	MΩ
Large Signal Voltage Gain	A_V	R_L≥2kΩ,V_O=±10V		86	100	-	dB
Maximum Output Voltage Swing 1	V_{OM1}	R_L≥10kΩ		± 12	± 14	-	V
Maximum Output Voltage Swing 2	V_{OM2}	R_L≥2kΩ		± 10	± 13	-	V
Input Common Mode Voltage Range	V_{ICM}			± 12	14	-	V
Common Mode Rejection Ratio	CMR	R_S≤10kΩ		70	90	-	dB
Supply Voltage Rejection Ratio	SVR	R_S≤10kΩ		76.5	90	-	dB
Operating Current	I_{CC}			-	3.5	5.7	mA
Slew Rate	SR			-	1	-	V/μs
Equivalent Input Noise Voltage (note2)	V_{NI}	RIAA,R_S=2.2kΩ,30kHz LPF	NJM4558x	-	1.4	-	μVrms
		RIAA,R_S=2.2kΩ,30kHz LPF	NJM4558x-D			1.8	
Gain Bandwidth Product	GB			-	3	-	MHz

(note2) In regard to Noise Standard, Nisshinbo Micro Devices is preparing for special D Rank type products (V_{NI}=1.8μVrms max.) except for SSOP package.

■ TYPICAL CHARACTERISTICS

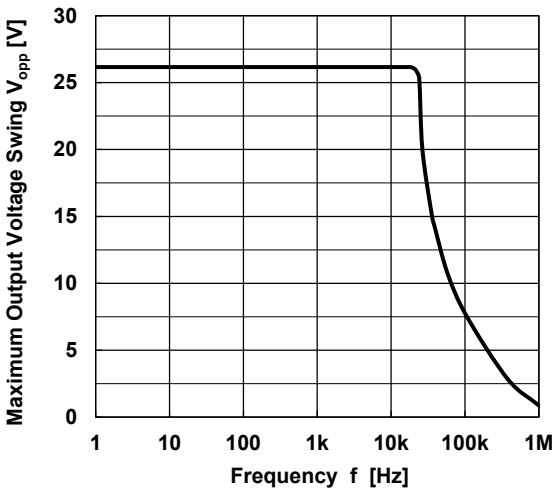
Open Loop Voltage Gain vs. Frequency

V^+/V^-=±15V, R_L=2kΩ, Ta=25ºC



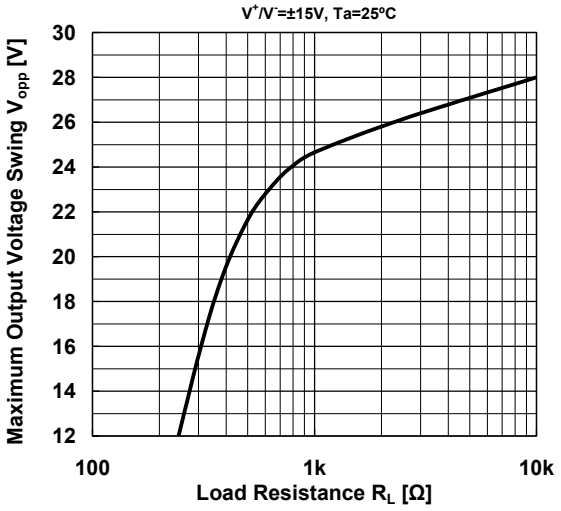
Maximum Output Voltage Swing vs. Frequency

V^+/V^-=±15V, R_L=2kΩ, Ta=25ºC



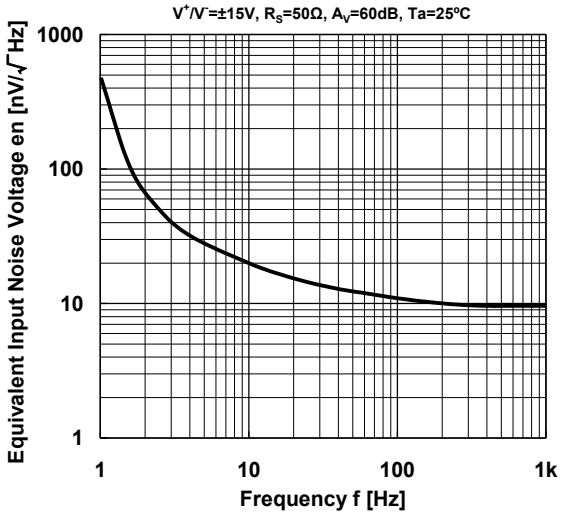
Maximum Output Voltage Swing vs. Load Resistance

V^+/V^-=±15V, Ta=25ºC



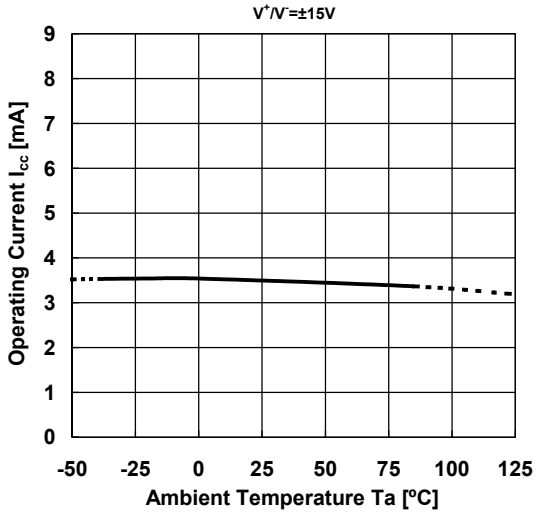
Equivalent Input Noise Voltage vs. Frequency

V^+/V^-=±15V, R_S=50Ω, A_V=60dB, Ta=25ºC



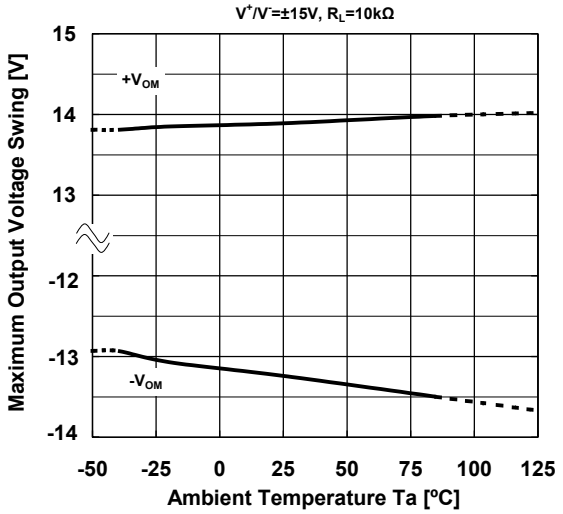
Operating Current vs. Temperature

V^+/V^-=±15V



Maximum Output Voltage Swing vs. Temperature

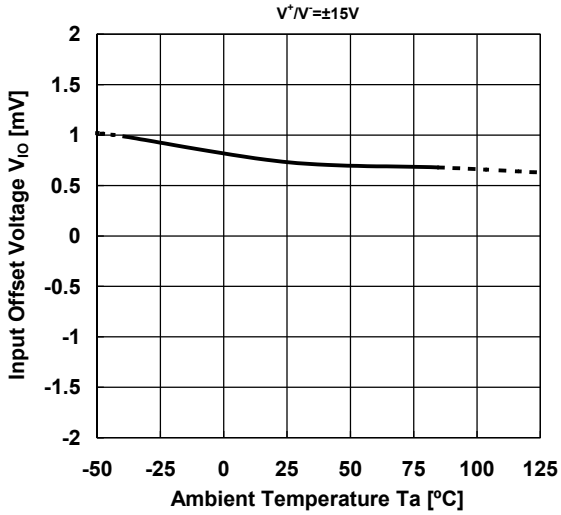
V^+/V^-=±15V, R_L=10kΩ



■ TYPICAL CHARACTERISTICS

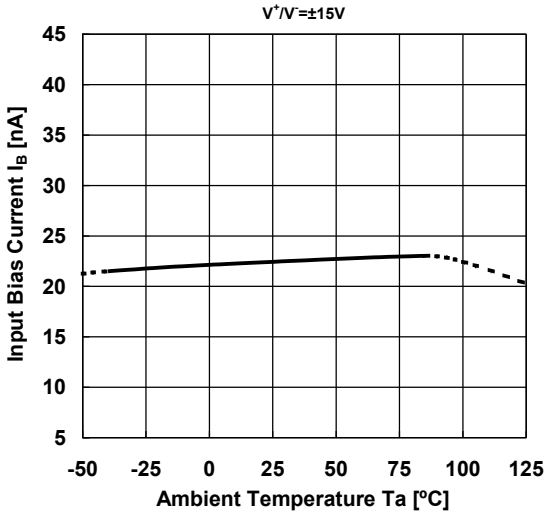
Input Offset Voltage vs. temperature

V^+/V^-=±15V



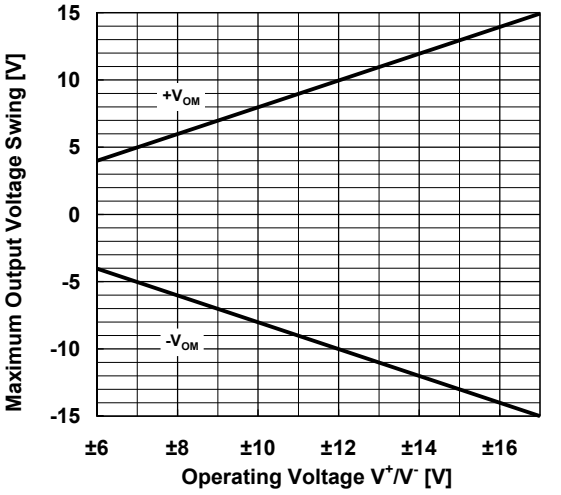
Input Bias Current vs. Temperature

V^+/V^-=±15V



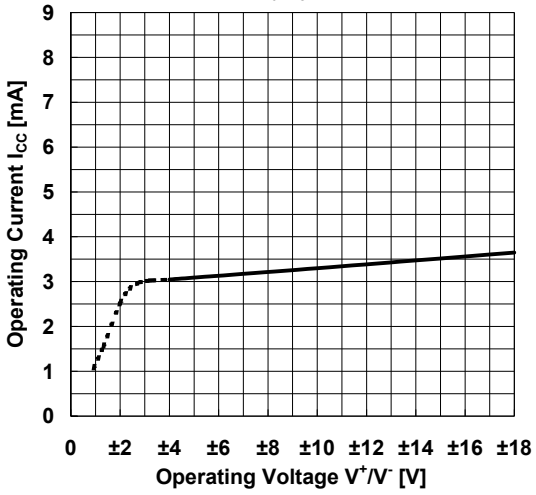
Maximum Output Voltage Swing vs. Operating Voltage

R_L=2kΩ, Ta=25ºC



Operating Current vs. Operating Voltage

Ta=25ºC



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